

DATABASE MANAGEMENT SYSTEM LAB

ASSIGNMENT NO. 03 – [Name : Ridafatema Patel, Rollno. 63]

-Consider the following database schemas :- (use all types of joins)

```
MariaDB [RIDA]> SELECT * FROM Student;
```

Stud_id	Stud_name	Instr_id	Stud_city
201	Aarati Kale	1	Pune
202	Rohit Patil	2	Nashik
203	Neha More	3	Mumbai
204	Akshay Jadhav	1	Pune
205	Pooja Deshmukh	4	Nagpur
206	Sagar Shinde	2	Nashik
207	Priya Pawar	3	Mumbai
208	Aditya Chavan	4	Pune
209	Kunal Patil	5	Aurangabad
210	Ishika Kulkarni	NULL	Pune

```
MariaDB [RIDA]> SELECT * FROM Instructor;
```

Instr_id	Instr_name	Instr_city	Specialization
1	Dnyaneshwar Pawar	Mumbai	Computer
2	Rida Patel	Nashik	AI & DS
3	Sahil Khan	Mumbai	Computer Networks
4	Karuna Shimpi	Pune	Data Science
5	Anjani Aute	Nagpur	Artificial Intelligence
6	Sidra Khan	Mumbai	Cyber Security
7	Hitesh Deore	Nashik	Software Engineering
8	Gaurav Pagare	Aurangabad	Web Development
9	Aditi Kale	Pune	Machine Learning
10	Om Dhole	Nashik	Cloud Computing
999	Guest Instructor	Pune	Mechanical

1. Find the instructor of each student.

```
MariaDB [RIDA]> SELECT s.Stud_id, s.Stud_name, i.Instr_name FROM Student s INNER JOIN Instructor i ON s.Instr_id = i.Instr_id;
```

Stud_id	Stud_name	Instr_name
201	Aarati Kale	Dnyaneshwar Pawar
202	Rohit Patil	Rida Patel
203	Neha More	Sahil Khan
204	Akshay Jadhav	Dnyaneshwar Pawar
205	Pooja Deshmukh	Karuna Shimpi
206	Sagar Shinde	Rida Patel
207	Priya Pawar	Sahil Khan
208	Aditya Chavan	Karuna Shimpi
209	Kunal Patil	Anjani Aute

2. Find the student who is not having any instructor.

```
MariaDB [RIDA]> SELECT s.Stud_id, s.Stud_name, i.Instr_name FROM Student s LEFT JOIN Instructor i ON s.Instr_id = i.Instr_id WHERE i.Instr_id IS NULL;
```

Stud_id	Stud_name	Instr_name
210	Ishika Kulkarni	NULL

3. Find the student who is not having any instructor as well as instructor who is not having student.

```
SELECT s.Stud_id, s.Stud_name, i.Instr_name FROM Student s LEFT JOIN Instructor i ON s.Instr_id = i.Instr_id UNION SELECT s.Stud_id, s.Stud_name, i.Instr_name FROM Student s RIGHT JOIN Instructor i ON s.Instr_id = i.Instr_id;
```

Stud_id	Stud_name	Instr_name
201	Aarati Kale	Dnyaneshwar Pawar
202	Rohit Patil	Rida Patel
203	Neha More	Sahil Khan
204	Akshay Jadhav	Dnyaneshwar Pawar
205	Pooja Deshmukh	Karuna Shimpi
206	Sagar Shinde	Rida Patel
207	Priya Pawar	Sahil Khan
208	Aditya Chavan	Karuna Shimpi
209	Kunal Patil	Anjani Aute
210	Ishika Kulkarni	NULL
NULL	NULL	Sidra Khan
NULL	NULL	Hitesh Deore
NULL	NULL	Gaurav Pagare
NULL	NULL	Aditi Kale
NULL	NULL	Om Dhole
NULL	NULL	Guest Instructor

4. Find the students whose instructor's specialization is computer.

```
SELECT s.Stud_name, i.Specialization FROM Student s JOIN Instructor i ON s.Instr_id = i.Instr_id WHERE i.Specialization = 'Computer';
```

Stud_name	Specialization
Aarati Kale	Computer
Akshay Jadhav	Computer

5. Create a view containing total number of students whose instructor belongs to Pune.

```
CREATE VIEW v1 AS SELECT COUNT(*) AS Total FROM Student s JOIN Instructor i ON s.Instr_id = i.Instr_id WHERE i.Instr_city = 'Pune';
```

```
MariaDB [RIDA]> CREATE VIEW v AS SELECT COUNT(*) AS Total FROM Student s JOIN Instructor i ON s.Instr_id = i.Instr_id WHERE i.Instr_city = 'Pune';
Query OK, 0 rows affected (0.007 sec)

MariaDB [RIDA]> SELECT * FROM v;
```

Total
2